

KEVIN E. WILK, PT, DPT, FAPTA^{1,2} • CHRISTOPHER A. ARRIGO, PT, MS, ATC^{3,4}
 MICHAEL S. BAGWELL, PT, DPT, CMPT¹ • MARCUS A. ROTHERMICH, MD⁵ • JEFFREY R. DUGAS, MD⁶

Repair of the Ulnar Collateral Ligament of the Elbow: Rehabilitation Following Internal Brace Surgery

Injury to the elbow in athletes who play overhead-throwing sports, especially baseball pitchers, continues to increase in frequency.¹⁰ The elbow of athletes in overhead-throwing sports is subject to severe and repetitive valgus stress.¹⁸ As a result, the ulnar collateral ligament (UCL), particularly the anterior band of the anterior oblique ligament, the primary soft tissue restraint to valgus elbow stress, is commonly injured.^{18,35} Chronic UCL insufficiency, leading to pain, medial elbow laxity, and inability to throw, results from microscopic ligamentous tears and attenuation.²⁴

Pitchers sustain most baseball-related injuries, accounting for 61% of all injuries in Major League Baseball, with 72% of these injuries affecting the throwing shoulder and elbow.¹⁰ Elbow injuries account for between 22% and 26% of all Major League Baseball pitching injuries.^{9,33} Conte et al¹⁰ reported that 25% of all Major League Baseball pitchers, and 14% of all minor-league pitchers on ac-

tive rosters, have undergone UCL surgery. Despite these statistics, the incidence of UCL injury and number of related surgeries continue to increase. Hodgins et al²¹ reported a 193% increase in UCL reconstruction surgery during the past 10 years in the state of New York. Petty et al³¹ reported a 50% increase in incidence of UCL reconstruction surgery in their cohort of high school baseball players. Ulnar collateral ligament injuries have been referred to as an epidemic in sports medicine.¹⁰

The greatest risk factors contributing to sustaining a pitching-related injury to the UCL in youth baseball pitchers are pitching when fatigued, a high pitch volume, improper mechanics, and repetitive throwing at maximal effort.²⁶ The greatest risk factor for a professional Major League Baseball pitcher is pitching at peak velocity, especially when combined with a large number of pitches.⁸

A recent advance in the surgical management of incomplete UCL tears in athletes is the technique of UCL repair with internal brace. This newly developed surgical approach and technique allows for faster rehabilitation and a quicker return to sports and other activities. The faster rehabilitation is allowed, because the collagen-treated internal brace is implanted inside the partially torn UCL to augment the ligament while it is healing. No surgical tunnels are needed, and there is no graft that requires time to heal and mature; therefore, there is no need for a prolonged recovery period.

The purpose of this commentary, based on our experience with more than 350 cases, including 79 patients with at least a 1-year postsurgical follow-up, was to describe and provide the rationale for the rehabilitation process we are using following UCL repair with internal brace, an emerging surgical procedure to treat selected UCL injuries.

• **SYNOPSIS:** Injuries to the elbow in athletes who play overhead sports, especially in baseball pitchers, continue to increase in frequency. The anterior band of the ulnar collateral ligament (UCL), the primary restraint to valgus stress, is commonly injured from throwing. Historically, such injuries have been treated with surgical reconstruction techniques, using a tendon autograft. A recently developed UCL repair procedure with an internal brace, utilizing collagen tape, is gaining popularity. The primary goal of this surgery is to enhance elbow joint stability while the ligament is healing and to allow earlier return to sport after

UCL reconstruction. The rehabilitation program following UCL repair with internal brace progresses through a different time frame than after UCL reconstruction. The purpose of this commentary, based on our experience with more than 350 cases, including 79 patients with at least a 1-year postsurgical follow-up, was to describe and provide the rationale for the rehabilitation process following UCL repair with internal brace. *J Orthop Sports Phys Ther* 2019;49(4):253-261. doi:10.2519/jospt.2019.8215

• **KEY WORDS:** baseball, elbow, pitching, surgery, throwing, UCL, ulnar collateral ligament

¹Champion Sports Medicine, Birmingham, AL. ²American Sports Medicine Institute, Birmingham, AL. ³Advanced Rehabilitation, Tampa, FL. ⁴MedStar Sports Medicine, MedStar Health at Lafayette Centre, Washington, DC. ⁵South Florida Orthopaedics & Sports Medicine, Stuart, FL. ⁶Andrews Sports Medicine & Orthopaedic Center, Birmingham, AL. No funding was provided for this article. The first 4 authors certify that they have no affiliations with or financial involvement in any organization or entity with a direct financial interest in the subject matter or materials discussed in the article. Dr Dugas is a paid consultant for Arthrex, Inc, and does receive a royalty on a kit put together to support the surgical procedure described in this article. The author does not receive royalties on the procedures performed or any specific components of the kit. Address correspondence to Dr Kevin Wilk, 805 St Vincent's Drive, Suite G100, Birmingham, AL 35205. E-mail: kwilkpt@hotmail.com • Copyright ©2019 *Journal of Orthopaedic & Sports Physical Therapy*®

UCL INJURY

THE INITIAL TREATMENT OF UCL INJURY is generally nonoperative^{35,36} and consists of rest from the overhead sport, anti-inflammatory medications, and rehabilitation. Rettig et al³⁵ reported that about half of the individuals with UCL injury can be treated successfully without surgery and are able to return to the same level of athletic activity. Surgery is indicated when symptoms do not abate or when symptoms recur after an appropriate trial of nonoperative treatment. Surgical treatment of a UCL injury should also be considered with acute UCL ruptures in high-level throwing athletes and for other athletes with significant chronic elbow laxity/instability, such as wrestlers, gymnasts, and cheerleaders.

Historically, the surgical treatment of choice for the UCL has been either reconstruction or a repair technique with primary sutures. In 1992, Conway et al,¹¹ after reviewing their results from both repair and reconstruction techniques, recommended reconstruction, based on their findings that overhead athletes performed significantly better post reconstruction compared to post repair. They suggested that repair should be considered only if the tear was a proximal avulsion, the procedure was performed soon after injury, the rest of the ligament was undamaged, and no symptoms attributed to the ulnar nerve were present.¹¹ Cain et al⁶ reported on the results of 743 UCL reconstructions with a minimum 2-year follow-up. They documented that 83% of the athletes who underwent reconstruction were able to return to the same or a higher level of play postoperatively. Accordingly, reconstruction surgery has become the treatment of choice for UCL deficiency, and there exists a large body of previously published work regarding the nonoperative rehabilitation of UCL injuries and rehabilitation post reconstruction.^{4,6,16,32,35,44,47,49,50} However, on average, 11.6 months are required before return to competition following UCL reconstruction surgery.⁶ Additionally, more

and more clinicians are suggesting a return to competition for pitchers after 18 months or longer.¹⁷

Based on recent advances in surgical technology, there is renewed interest in repair techniques for the UCL, particularly a surgical procedure referred to as the “UCL repair with internal brace” (*InternalBrace*; Arthrex, Inc, Naples, FL). The technique was developed, in part, based on the work of Conway et al,¹¹ along with a desire to have a surgical option that would allow a faster recovery than that required after UCL reconstruction.

Ulnar collateral ligament repair with internal brace technique is reserved for specific types of UCL injury.¹⁵ These include partial or complete tears at the origin or distal insertion of the UCL with good ligament tissue and low-grade, mid-substance partial UCL tears. Within these parameters, surgical repair of the UCL with an internal brace can be performed in the presence of open physes.¹⁵ In patients with chronic attritional damage to the UCL and associated loss of elbow joint stability, reconstruction remains the most appropriate surgical intervention.^{4,6,11,22} Therefore, injured athletes with good joint stability and high-quality native ligament tissue are the ideal candidates for UCL repair with internal brace.

SURGICAL PROCEDURE

THE SURGICAL TECHNIQUE FOR UCL repair with internal brace was described in 2016 and includes augmentation of the native ligament repair with a spanning tape dipped in collagen (*InternalBrace*) and anchored on each end of the native UCL.¹⁵ Surgical dissection of the medial elbow is identical to the standard dissection for UCL reconstruction, with elevation of the flexor pronator mass to expose the native ligament (FIGURE 1). Importantly, all patients are counseled prior to the procedure that the decision to proceed with surgical repair of the UCL, rather than reconstruction, can only be made intraoperatively from a direct visual assessment of the native ligament following proper exposure. In patients with acceptable-quality native tissue, the decision is then made to perform the UCL repair with internal brace or tape augmentation.

At that point, the tear is identified as a proximal injury at the medial epicondyle of the humerus or a distal injury at the sublime tubercle of the ulna. Drill holes are made at both the proximal and distal footprints of the native UCL. A 3.5-mm SwiveLock (Arthrex, Inc) loaded with a 2-mm FiberTape (Arthrex, Inc) and a size-0 nonabsorbable suture are anchored at the end of the injury (proximal

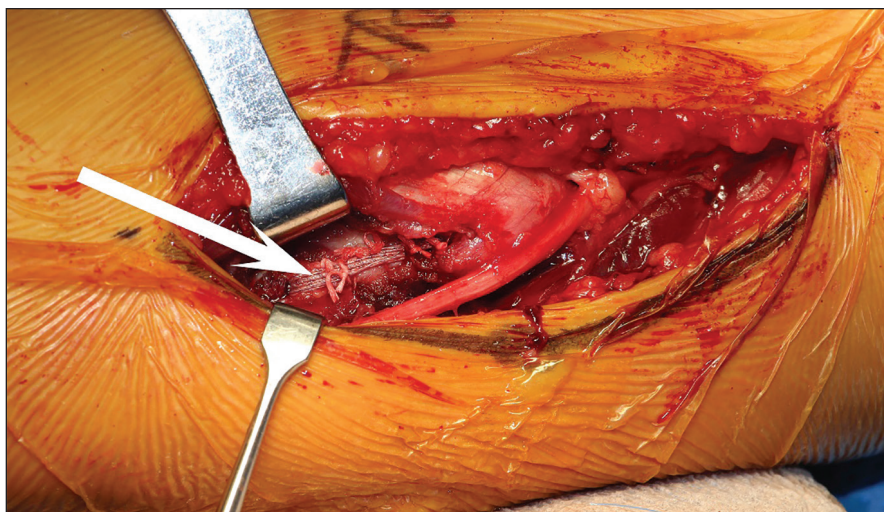


FIGURE 1. Ulnar collateral ligament repair with an internal brace inserted within the ligament. The arrow indicates the exact location of the internal brace.

or distal). The native ligament is then repaired to its origin or insertion with the nonabsorbable suture, and the longitudinal split in the flexor pronator mass tendon is repaired with 3 additional size-2-0 sutures.

A second 3.5-mm SwiveLock anchor is then placed at the other end of the native UCL and advanced such that the tension of the FiberTape matches the native UCL during elbow range of motion (ROM). This second SwiveLock anchor is placed with the elbow in slight flexion and varus, and 3 additional size-0 nonabsorbable sutures are passed around the repaired native ligament and FiberTape augmentation for incorporation. The ulnar nerve is only transferred in individuals who were experiencing ulnar nerve symptoms prior to surgery. When the ulnar nerve is transferred, it is done with a subcutaneous transposition.

Dugas et al¹⁵ reported on the biomechanical analysis comparing gap formation upon valgus stress and maximal torque to failure of the augmented UCL repair with internal brace and the typical modified Jobe UCL reconstruction technique. They demonstrated a significantly greater resistance to gapping with valgus stress with the UCL repair with internal brace (mean $\pm$ SD, 0.35 ± 0.16 mm) compared to UCL reconstruction (0.53 ± 0.23 mm).¹⁵ The results of this study suggest that the technique of augmented UCL repair with internal brace replicates failure strength of the traditional UCL reconstruction and provides greater elbow stability at low cyclic loads. When extrapolated to patients undergoing elbow surgery at a young age, this may represent a viable treatment option for long-term elbow stability.

POSTSURGICAL REHABILITATION

THE PROPOSED REHABILITATION AFTER UCL repair with internal brace follows a sequential and progressive multiphased approach.^{27,47,48} The program incorporates a combination of time- and

criterion-based approaches to account for soft tissue physiologic healing requirements and individual patient progression. The ultimate goal of the rehabilitation program is to return the athlete to his or her previous functional level, or better, as quickly and safely as possible. The program detailed in **APPENDIX A** (available at www.jospt.org) consists of 5 phases, with the eventual goal of a gradual return to competitive throwing 5 months following the surgery.

Phase 1: Immediate Postoperative Phase

The initial phase of the rehabilitation process is intended to protect healing tissue, reduce pain and inflammation, retard muscular atrophy, and regain full wrist/shoulder motion, while allowing early healing of the surgical repair. Therefore, for the first 7 days after the surgery, the patient is placed in an adjustable ROM elbow brace, with the elbow immobilized at 90° of flexion to protect the healing UCL. No elbow or forearm mobility exercises are performed. However, the patient is allowed wrist and hand movement. This also allows adequate healing of the soft tissue slings involved when an ulnar nerve transposition is performed.⁴

In this phase, grade I and II mobilization techniques to the humeroulnar and humeroradial joints are used to neuromodulate pain by stimulating type I and II articular receptors.^{23,53} Cryotherapy and high-voltage electrical stimulation may be performed to further assist in reducing pain and inflammation. We also use class IV deep tissue laser over the incision, with the intent to speed healing and increase nitrous oxide levels in the healing tissue.⁵ Beginning on days 14 to 17 postoperative, once the acute inflammatory response related to the surgical procedure has subsided, moist heat may be used at the onset of treatment to prepare the tissue for stretching and improve the extensibility of the capsule and musculotendinous structures.⁴⁷

The early phase of rehabilitation also focuses on voluntary activation of muscle to help minimize muscular at-

rophy, with all exercises performed in the elbow brace locked at 90° of flexion. Nonpainful, submaximal isometrics are performed initially for the elbow flexors and extensors, as well as for the wrist flexor, extensor, pronator, and supinator muscle groups. Shoulder isometrics for external rotation (ER), internal rotation (IR), flexion, extension, and abduction may also be performed, with the shoulder adducted to the side and in neutral rotation, to minimize the strain on the UCL repair. Care must be taken to avoid ER and IR exercises if they are painful. Alternating rhythmic stabilization drills for shoulder ER/IR and flexion/extension and for elbow flexion, extension, supination, and pronation are performed to begin re-establishing proprioception and neuromuscular control of the upper extremity.

Additionally, seated scapular manual resistance and postural exercises are introduced along with postural stretching to improve the patient's scapular position and posture. Following the initial 7-day immobilization period to ensure adequate fascial sling healing after surgery, trunk, core, pelvic girdle, and lower extremity exercises can be incorporated into the rehabilitation program.

Phase 2: Controlled Mobility Phase

The controlled mobility phase is initiated at the beginning of postoperative week 2 and ends after week 5. This phase of rehabilitation focuses on gradually restoring elbow joint ROM, improving muscular strength and endurance, and normalizing joint arthrokinematics. On day 8 post surgery, the elbow ROM brace is set to allow elbow motion from 30° to 110° of flexion. The setting is adjusted again to allow motion from 10° to 125° at the beginning of week 3. Finally, at the beginning of week 4, the setting is progressed to allow full elbow ROM (0°-145°). The brace is discontinued by week 6, during the transition from the controlled mobility to the intermediate phase. With this protocol, the restoration of full ROM is faster than that seen with UCL re-

construction surgery for many reasons, including less postoperative pain and associated muscle guarding, no osseous tunnels performed during surgery, and no secondary pain from graft harvesting.

In this phase, controlled passive elbow and forearm ROM exercises are performed by the rehabilitation specialist to promote joint nutrition of the articular cartilage and assist in the synthesis, alignment, and organization of collagen tissue.^{12,13,28,30,36-39} Active-assisted, active, and passive ROM exercises are performed for the elbow, forearm, and wrist to restore elbow flexion and extension, forearm supination and pronation, and wrist motion, as well as to prevent the formation of scar tissue and adhesions.⁴⁵

A primary goal of early ROM activities is to minimize the occurrence of elbow flexion contractures and achieve full elbow extension or preinjury motion.^{1,19,44} Accordingly, preoperative elbow motion needs to be assessed and recorded. Patients should be asked whether they have had full elbow extension in the past 2 to 3 years. Postoperative ROM often is related to preoperative motion, especially with UCL repair in athletes who throw. Intraoperative ROM should also be recorded. Elbow flexion contractures can be a deleterious consequence of surgery for the overhead athlete, as the elbow has a predisposition to flexion contractures because of the intimate congruency of the joint articulations, joint capsule tightness, and tendency of the anterior capsule to develop adhesions after injury.⁴⁰ Additionally, the brachialis muscle has a capsular attachment as it crosses the elbow joint prior to becoming a tendinous structure. Injury to the elbow can cause excessive scar tissue formation in the brachialis muscle and adjacent tissues. In this instance, functional splinting of the elbow may be indicated.^{40,47,50}

In addition to ROM exercises, joint mobilizations are performed as indicated by accessory mobility assessment to address all capsular restrictions evident in

the elbow, shoulder, and/or wrist to minimize the occurrence of joint contractures. Posterior glides with oscillations are performed at end range of available motion to help regain full elbow extension. Grade I and II mobilizations are used initially, with a progression to more aggressive mobilization techniques (grades III and IV) as the rehabilitation progresses into the later stages, when symptoms have subsided. Joint mobilization should also be performed for the radiohumeral and radioulnar joints.^{47,50}

If the patient continues to have difficulty achieving full extension using ROM and mobilization techniques, a low-load, long-duration stretch (also called collagen creep) may be performed to produce a deformation or creep of the collagen tissue, resulting in tissue elongation.^{19,25,27,39} We have found this technique to be extremely beneficial for regaining full elbow extension. The patient lies supine, with a towel or foam roll placed under the distal humerus to act as a cushion and fulcrum. A light-resistance elastic exercise band is applied to the wrist and secured to the table or to a dumbbell on the ground (**FIGURE 2**). The patient is instructed to relax as much as possible for 12 to 15 minutes in this elbow-extended (elongated) position. The patient should perform this exercise at home periodically during the day, totaling 60 minutes of time under tension per day. The applied force should be at a level tolerable to allow the performance of the stretch for the entire time without experiencing pain or muscle guarding, thereby imparting a low-load, long-duration stretch.^{47,50}

In addition to restoration of ROM, at the beginning of week 3, the patient is instructed to begin active, concentric shoulder- and elbow-strengthening exercises. Dynamic joint stability at the elbow is created by compressive forces produced by the muscles crossing the joint, particularly those of the flexor-pronator group medially.²⁰ Isotonic and manual resistive activities for the flexor-pronators as a whole, and the flexor carpi ulnaris and

flexor digitorum superficialis specifically, are performed to generate compressive forces across the elbow joint in a gradually progressive fashion. Based on their anatomical orientation overlying the UCL, these muscles assist in stabilizing valgus stress at the medial elbow. These muscles have also been shown to have significantly decreased electromyographical signal intensity in pitchers with UCL deficiency, indicating a need for rehabilitation.²⁰ Strengthening exercises are initiated using concentric exercises and progressed to eccentric muscle contractions. We suggest that the rehabilitation program also emphasize developing a comprehensive strengthening program for the throwing athlete, using the previously published Throwers Ten program,^{42,43} which is summarized in **APPENDIX B** (available at www.jospt.org). The Throwers Ten program is initiated at the beginning of week 3 or 4, based on the patient's progression and signs and symptoms.

Phase 3: Intermediate Phase

This phase is initiated in the beginning of week 6 and progresses to week 8. The emphasis in this phase includes maintaining (or fully restoring, if still limited) elbow and upper extremity mobility, improving muscular strength and endurance, re-establishing neuromuscular control of the elbow complex, and continuing with a functional progression of activity.

Stretching exercises are continued, as needed, to maintain full elbow, forearm, and wrist ROM. Mobilization techniques should be progressed, as required, to address any remaining capsular or accessory motion limitations. The aggressiveness of stretching and mobilization techniques should be matched with the healing constraints and time frames of the involved tissues, the specific pathology or surgical technique, and the amount of motion and perceived end feel at the end of motion. If the patient has decreased motion and an abnormally hard end feel in the absence of pain, aggressive stretching and mobilization techniques may be indicated. Conversely, a patient who experiences

pain before resistance or presents with an empty end feel must undergo a slower progression, with gentle mobilization and stretching.

Flexibility is progressed during this phase, with focus on wrist flexors and extensors as well as the forearm pronators and supinators. Elbow extension and forearm pronation flexibility is specifically emphasized in throwing athletes so that they can perform efficiently. It is also important to maintain optimal shoulder flexibility in overhead athletes. Even a slight loss of shoulder ER at 90° of abduction may result in increased strain on the medial elbow structures during the overhead throwing motion. Stretching to achieve and maintain optimal shoulder IR ROM at 90° of abduction, flexion, and horizontal adduction should also be diligently performed.^{45,47,50} In the overhead athlete, it is important to assess shoulder total rotational ROM and normalize motion to the opposite extremity.^{48,49}

Neuromuscular control exercises of the shoulder and elbow complexes are advanced in this phase to enhance the muscles' ability to control the elbow joint during athletic activities. Neuromuscular control is the unconscious trained response of a muscle to a signal regarding dynamic joint stability.⁵⁴ This system of neurological messages is sometimes referred to as "muscle memory." Throwing requires strength, endurance, and precision. Training the neurologic system requires that the movement be performed correctly on a repetitive basis.⁵⁴



FIGURE 2. Low-load, long-duration stretching to improve elbow extension. A low-intensity stretch is applied for 10 to 12 minutes. Note that the forearm is pronated and the shoulder internally rotated to lock the humerus into a position that minimizes compensation.

Constantly repeated neuromuscular control drills are performed against manual resistance to allow the clinician to perform a continual assessment of the athlete's progression regarding strength and control, as well as to resist the athlete through combined motions of the elbow and wrist. These exercises include proprioceptive neuromuscular facilitation exercises with rhythmic stabilizations and slow-reversal manual resistance for elbow and wrist flexion (**FIGURE 3**; **ONLINE VIDEO**, available at www.jospt.org), along with resisted scapular protraction, retraction, elevation, and depression. While emphasis is placed on developing dynamic stabilization of the medial elbow, it is important to train the scapular musculature to minimize postsurgical strength loss and address any pre-existing deficits in dynamic control.^{45,47,50} In addition to traditional shoulder IR and ER strengthening exercises with elastic resistance bands, we commonly progress these exercises to include superimposed manual rhythmic stabilizations in standing (**FIGURE 4**). Then, to further increase the difficulty of the exercise on the shoulder and include trunk/hip stabilization, the exercise is performed on a stability ball (**ONLINE VIDEO**, available at www.jospt.org).

Strengthening exercises are progressed at weeks 6 to 8 to the previously published Advanced Throwers Ten exercise program,⁵² which is summarized in **APPENDIX C** (available at www.jospt.org). With this program, the entire arm, in particular the glenohumeral and scapu-



FIGURE 3. Proprioceptive neuromuscular facilitation manual resistance during combined elbow flexion, wrist flexion, and forearm pronation to enhance dynamic elbow stability (**ONLINE VIDEO**, available at www.jospt.org).

lothoracic muscles, has a higher demand placed on it, to emphasize strengthening of the shoulder external rotators and scapular muscles in preparation for the forces of throwing.⁵² These exercises incorporate high-level endurance and alternating movement patterns to further challenge shoulder girdle neuromuscular control and facilitate the rotator cuff musculature via alternating dynamic movements with the addition of sustained contralateral isometric holds.⁷

The incorporation of sustained holds challenges the athlete to maintain a set position while the opposite extremity performs superimposed isotonic movements. Two sets are incorporated into each exercise, each following a sequence integrating bilateral isotonic movement and unilateral isotonic movement with contralateral sustained holds. The athlete performs these exercises in standing and then progresses to a seated or prone position on a stability ball (depending on the specific exercise). This creates an unstable surface that further challenges the trunk and posterior hip musculature to maintain a stable base of support for exercise performance. Manual resistance provided by the clinician is employed with the



FIGURE 4. Standing external rotation with elastic resistance at 0° of abduction, with manual rhythmic stabilization performed at end range (**ONLINE VIDEO**, available at www.jospt.org).

seated stability ball exercises to augment muscle excitation and improve endurance of the shoulder, arm, and trunk/hip musculature (**FIGURE 5**; **ONLINE VIDEO**, available at www.jospt.org). A complete description of this program has previously been published.⁵²

The initiation of an upper extremity, 2-hand plyometric program⁵¹ is done at week 6, and a 1-hand plyometric program⁵¹ is initiated at week 8. Plyometric drills can be an extremely beneficial form of functional exercise for training the elbow in overhead athletes.^{3,45,47,48,50} Plyometric exercises are performed using a weighted medicine ball during the later stages of phase 3 to train the shoulder and elbow to develop and withstand high levels of stress. Two-hand plyometric exercises include chest passes, side-to-side throws, and overhead soccer throws. These are progressed to 1-hand activities, such as wall dribbles, 90°/90° throws against the wall (**FIGURE 6**), ER and IR throws at 0° of abduction, and throws against the rebounder (**FIGURE 7**). Important specific plyometric drills that emphasize the forearm and hand musculature include wrist flexion flips (**FIGURE 8**; **ONLINE VIDEO**, available at www.jospt.org) and extension grips.^{45,47,50}

Phase 4: Advanced Strengthening Phase

The fourth phase of rehabilitation involves a progression of activities to prepare the athlete for high-level stress situations and participation in sports.



FIGURE 5. Seated on a stability ball, the athlete performs shoulder external rotation using elastic-band resistance and manual resistance (**ONLINE VIDEO**, available at www.jospt.org).

The time frame for this phase is from week 9 to week 14. The goals of this phase are to gradually increase strength, power, endurance, and neuromuscular control to prepare for a gradual, progressive return to sports. Before initiating this phase, the following criteria must be met: full, nonpainful ROM; absence of pain or tenderness to palpation; a satisfactory clinical exam that includes no pain or laxity, specifically with valgus stress²⁹ and milking maneuvers,⁴¹ but also with any other special test for elbow pathology; completion of prior rehabilitation phases without difficulty; and muscle strength that meets the criteria to throw. We have established and previously published return-to-throwing criteria based on objective clinical testing that the patient must fulfill prior to initiating an interval throwing program (**APPENDIX D**, available at www.jospt.org).^{14,46,48,49}

Aggressive exercises involving eccentric and plyometric movements are continued in this phase. Advanced strengthening activities emphasize high speed, eccentric contractions, and plyometric activities. Elbow flexion exercises are progressed to emphasize eccentric control. The biceps muscle is an important stabilizer during the follow-through



FIGURE 6. Plyometric throwing drill: in standing, the athlete throws a 0.9-kg (2-lb) plyoball into the wall to perform a stretch-shortening contraction.

phase of overhead throwing; it eccentrically controls the deceleration of the elbow, preventing forceful abutment of the olecranon within the fossa.^{3,18} Elbow flexion exercises can be performed with elastic tubing to emphasize slow and fast concentric and eccentric contractions. Aggressive strengthening exercises performed with weight machines are also incorporated during this phase at week 10. This is most commonly done with a seated chest press, seated rowing, and front latissimus dorsi pull-downs in addition to the Advanced Throwers Ten exercise program (**APPENDIX C**). The triceps is exercised primarily using concentric exercises because of the acceleration (muscle shortening) activity of the triceps during the acceleration phase of throwing.¹⁸

Neuromuscular control exercises are progressed again to include sidelying shoulder ER with increased manual resistance and repetitions. Concentric and eccentric shoulder ER exercises are performed against the clinician's resistance with the addition of rhythmic stabilizations.^{45,47,50}

Additional challenging exercises for the higher-level athlete include side planks with superimposed shoulder ER strengthening and lateral slides with shoulder ER. The goal of these exercises is to improve core strength and further challenge the posterior shoulder musculature in preparation for a return to throwing.

An interval hitting program is permitted at week 10, while an interval



FIGURE 7. Plyometric throwing drill: in standing, the athlete throws a 0.9-kg (2-lb) plyoball into a rebounder at 90°/90° of abduction.

throwing program for the overhead athlete is allowed at week 12 if the athlete is ready.^{45,47,50} Reinold et al³⁴ previously detailed a variety of interval programs, including both interval hitting and throwing programs for the baseball athlete. The interval throwing program begins at 14 m (45 ft) and gradually progresses to 37 and then to 55 m (120 and 180 ft), based on the player and playing position.³⁴ Throwing should be performed without pain or any significant increase in symptoms at every step in the progression. Prior to throwing, it is important for the overhead athlete to perform stretching and an abbreviated strengthening program. Throwers are instructed to warm up (a clinician may aid with stretching), and then they perform 1 set of their exercise program prior to throwing, followed by 2 additional sets of exercises after throwing.³⁴ This not only provides an adequate warm-up, but also ensures maintenance of necessary ROM and flexibility of the shoulder and elbow.

In most cases, pitchers will progress to throwing from a mound approximately 8 weeks after initiation of an interval throwing program.³⁴ Generally, the pitcher is started at 50% perceived intensity and gradually progressed to 75%, 90%, and 100%. The actual rate of progression of the throwing program should be individualized to each athlete and adjusted based on the presence of any symptoms, mechanics, and desired goals. For a detailed interval program and progression guidelines, we refer readers to the previously published paper by Reinold et al.³⁴

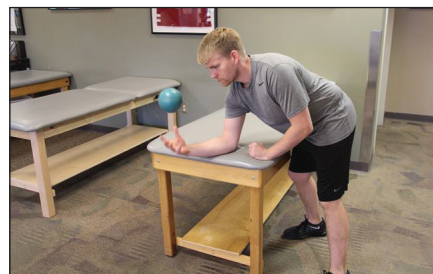


FIGURE 8. Wrist flexion ball flips performed with a 0.9 kg (2 lb) plyoball (ONLINE VIDEO, available at www.jospt.org).

Phase 5: Return-to-Activity Phase

The final phase is the return-to-activity phase. During this phase, proper dynamic warm-up is reinforced and exercise progression is continued, allowing the athlete to progressively return to full activity and competitive throwing after completing an interval athletic program that could be for throwing, tennis, golf, swimming, volleyball, etc.³⁴ We have established a set of return-to-play criteria for the overhead throwing athlete. These criteria include 6 areas of evaluation, including 4 functional test measures (APPENDIX D).¹⁴

Upper extremity functional tests can help inform the decision on when the athlete is ready to return to play. A very useful test for the throwing athlete, developed by the senior author (K.W.), is the prone ball-drop test. This test is conducted with the patient in prone and the arm to be evaluated off the side of a treatment table. The shoulder is abducted to 90° and the elbow fully extended off the table, with the palm of the hand facing down toward the floor. In this position, the athlete holds a 0.9 kg (2 lb) plyoball. From this position, the athlete quickly releases and catches the ball as many times as possible in 30 seconds (FIGURE 9; ONLINE VIDEOS, available at www.jospt.org). The total number of successful catches is compared bilaterally, with the goal for the throwing arm of 110% of the nonthrowing side.

The final decision for return to competition, when the athlete has completed the rehabilitation program and met the return-to-throwing criteria specified in APPENDIX D, is made by the physician and the physical therapist or rehabilitation team. In our experience to date, the

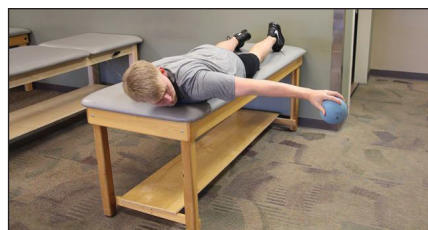


FIGURE 9. Positioning for prone ball-drop testing (ONLINE VIDEO, available at www.jospt.org).

typical time frame to return to play following this procedure has been approximately 5 months after surgery. This is in sharp contrast with the reported average return-to-competition timeline of 11.6 months following UCL reconstruction.⁶ Some authors have recommended 18 to 20 months of rehabilitation prior to return to competition.¹⁷

Comparison With UCL Reconstruction

For both UCL reconstruction and UCL repair with internal brace, elbow motion is restricted and held in 90° of flexion for 7 days following surgery, with the goal to protect the healing UCL, and fascial slings are used in instances of concomitant ulnar nerve transposition. Following this first week, elbow motion is generally progressed faster after UCL repair with internal brace, with extension allowed to tolerance and full ROM expected by the end of week 4. This is in contrast to the expectation of full elbow ROM by the end of week 6 after UCL reconstruction. The Throwers Ten exercise program is initiated 3 or 4 weeks following surgery, and the Advanced Throwers Ten program 6 to 8 weeks post surgery, in patients with a UCL repair with internal brace. This is in contrast to week 6 and weeks 8 to 9, respectively, for patients with a UCL reconstruction. Compared to UCL reconstruction, following UCL repair with internal brace, 2-handed plyometric drills are started 2 weeks sooner (6 weeks post surgery) and 1-handed drills 8 weeks sooner (8 weeks post surgery). Isotonic machine resistance exercises are begun at 10 weeks, 2 to 4 weeks earlier than when they are initiated following UCL reconstruction.

A significant difference between the rehabilitation programs post UCL repair with internal brace compared to UCL reconstruction is the much earlier initiation of interval training programs and return to competitive throwing. With UCL repair with internal brace, interval hitting and golf can be incorporated 10 weeks post surgery, which is 2 to 4 weeks sooner than following UCL reconstruction. Long

toss (interval throwing phase 1) begins at 12 weeks (compared to 16 weeks) post surgery, and throwing off the mound (interval throwing phase 2) is initiated 16 to 20 weeks (compared to 22-24 weeks) post surgery. Gradual return to competitive throwing begins 5 months following UCL repair with internal brace, in contrast to 9 to 12 months following UCL reconstruction.^{6,17}

SUMMARY

THE UCL IS A FREQUENTLY INJURED structure in overhead throwing athletes, with an increasing incidence of injury in the young athlete.^{8-10,17,21,31,33} There has been renewed interest in performing surgical repair of the UCL in athletes who meet specific injury criteria. In this commentary, we outlined our postsurgical rehabilitation approach for UCL repair with internal brace, based on treating more than 350 athletes who have undergone this procedure over the past 2 years. The amount of time required to return to play after UCL repair with internal brace has been approximately 5 months, nearly 7 months faster than following UCL reconstruction surgery.⁶ While the purpose of this commentary was not to present full outcome data, the initial follow-up findings following UCL repair with internal brace are promising. Further research is needed to determine the long-term results of this surgery and the effectiveness of the rehabilitation program described.

Surgical and rehabilitative effectiveness is ultimately gauged by the successful return to preinjury level of competition in the athletic population. Therefore, the key long-term outcome measures needed to evaluate the viability and efficacy of surgical repair of the UCL with internal brace and rehabilitation should include, at a minimum, return-to-competition follow-up of greater than 2 years coupled with Kerlan-Jobe Orthopaedic Clinic shoulder and elbow functional assessment scores of greater than 85.^{2,14} ●

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APPENDIX A

POSTOPERATIVE REHABILITATION FOR ULNAR COLLATERAL LIGAMENT REPAIR WITH INTERNAL BRACE

Phase 1: Immediate Postoperative Phase (Week 1)

Goals: protect healing tissue, reduce pain and inflammation, retard muscle atrophy, achieve full wrist ROM

Day of Surgery

1. Elbow brace locked at 90° for 7 days
2. PROM of wrist and hand in locked brace

Postoperative Days 1 and 2: Add (all performed in locked elbow brace)

1. Shoulder PROM: flexion, ER, and IR to tolerance
2. Shoulder pendulum exercises
3. Wrist flexor/extensor stretching
4. Putty/gripping exercises

Postoperative Days 3 Through 7 (all exercises performed in locked elbow brace)

1. Continue previous exercises, advancing PROM as tolerated
2. Add the following exercises:
 - a. Shoulder isometrics: ER, IR, abduction, flexion, and extension performed pain free, submaximal
 - b. Scapular strengthening: seated neuromuscular control drills with manual resistance

Progression to next phase is purely time based

Phase 2: Controlled Mobility Phase (Weeks 2-5)

Goals: gradually restore elbow ROM, improve muscular strength and endurance, normalize joint arthrokinematics

Beginning Week 2 (Day 8)

1. Set elbow ROM brace to 30° to 110°
2. Begin elbow PROM and active-assisted ROM from 30° to 110°
3. Initiate elbow AROM for flexion
4. Initiate shoulder AROM in elbow brace
5. Progress scapular-strengthening exercises
 - a. Seated manual resistance: protraction/retraction, elevation/depression, diagonal patterns
6. Progress to light isotonic strengthening exercises for wrist, elbow, and shoulder at day 10

Beginning Week 3

1. Progress elbow ROM to 10° to 125°
2. Initiate Throwers Ten exercise program

Beginning Week 4

1. Progress elbow ROM to 0° to 145°
2. Progress elbow- and wrist-strengthening exercises
3. Initiate wrist flexion and elbow flexion movements against manual resistance

Criteria for progression to next phase: elbow PROM of 10° to 125°, minimal pain and tenderness, good manual muscle testing of key muscles/movements (elbow flexion/extension; wrist flexion; shoulder IR, ER, and scapular abduction)

Phase 3: Intermediate Phase (Weeks 6-8)

Goals: restore full elbow ROM, progress upper extremity strength, continue with functional progression

Beginning Week 6

1. Discontinue elbow brace at week 6
2. Initiate Advanced Throwers Ten program
3. Initiate 2-hand plyometrics: chest pass, side-to-side throw, and overhead pass
4. Initiate prone plank exercise (**ONLINE VIDEO**)

APPENDIX A

Beginning Week 8

1. Progress to 1-hand plyometrics: 90°/90° ball throw, 0° ball throw
2. Continue with Advanced Throwers Ten program
3. Initiate side plank with shoulder ER strengthening exercise

Criteria for progression to next phase: full, nonpainful elbow AROM and PROM; no pain or tenderness; appropriate strength of shoulder and elbow (minimum of 70% compared to opposite side); satisfactory clinical examination; completion of current rehabilitation phase without difficulty

Phase 4: Advanced Phase (Weeks 9-14)

Goals: advanced strengthening exercises, initiate ITP, gradual return to throwing

Beginning Week 9

1. Continue all strengthening exercises, including Advanced Throwers Ten program and 1- and 2-hand plyometrics program

Beginning Week 10: Initiate

1. Seated chest-press machine
2. Seated row machine
3. Biceps/triceps machine or cable strengthening
4. Interval hitting program

Week 12: Initiate ITP Phase 1

1. Long toss

Criteria to enter next phase: full elbow, wrist, and shoulder ROM; no pain or tenderness; functional or isokinetic test that fulfills criteria for desired activity; satisfactory clinical examination

Phase 5: Return-to-Activity/Play Phase (Weeks 14+)

Goals: gradual return to competitive throwing, continue all exercises and stretches

Weeks 14 to 16

1. Continue all exercises as in weeks 9 to 14
2. Continue/progress ITP phase 1 (long toss)
 - a. Each athlete may progress through ITP at different rates/paces
 - b. Expected to complete throws of 0 to 27 m (0-90 ft) within 3 weeks of starting ITP and throws of 0 to 37 m (120 ft) within 8 weeks

Weeks 16 to 20

1. Continue ROM and stretching programs
2. Continue Advanced Throwers Ten program
3. Continue plyometrics
4. Initiate ITP phase 2 (off the mound) when phase 1 is complete and athlete is ready

Weeks 20+

1. Initiate gradual return to competitive throwing
2. Perform dynamic warm-ups and stretches
3. Continue Advanced Throwers Ten program
4. Return to competition when athlete is ready (physician decision and rehabilitation team)

Abbreviations: AROM, active range of motion; ER, external rotation; IR, internal rotation; ITP, interval throwing program; PROM, passive range of motion; ROM, range of motion.

APPENDIX B

EXERCISES INCLUDED IN THE THROWERS TEN EXERCISE PROGRAM

- Diagonal-pattern D2 extension
- Diagonal-pattern D2 flexion
- Shoulder external rotation at 0° of abduction
- Shoulder internal rotation at 0° of abduction
- Shoulder abduction to 90°
- Shoulder scapular abduction, external rotation ("full cans")
- Sidelying shoulder external rotation
- Prone shoulder horizontal abduction
- Prone shoulder horizontal abduction (full external rotation, 100° of abduction)
- Prone rowing
- Prone rowing into external rotation
- Press-ups
- Push-ups
- Elbow flexion
- Elbow extension
- Wrist extension
- Wrist flexion
- Wrist supination
- Wrist pronation

All exercises are performed against resistance, with the goal to improve strength. Full description of the program is available.^{45,46}

APPENDIX C

EXERCISES INCLUDED IN THE ADVANCED THROWERS TEN EXERCISE PROGRAM

Elastic Tubing/Band Resistive Exercises

- Shoulder external rotation at 0° of abduction while seated on a stability ball*
- Shoulder internal rotation at 0° of abduction while seated on a stability ball*
- Shoulder extensions while seated on a stability ball†
- Lower trapezius isolation while seated on a stability ball†
- High row into shoulder external rotation while seated on a stability ball†
- Biceps curls/triceps extensions while seated on a stability ball†

Isotonic Dumbbell Resistive Exercises

- Full can while seated on a stability ball†
- Lateral raise to 90° while seated on a stability ball†
- Prone T's on stability ball†
- Prone Y's on stability ball†
- Prone row into external rotation on stability ball†
- Sidelying shoulder external rotation
- Wrist flexion/extension and supination/pronation

*Contralateral sustained hold performed during exercise.

†Exercises are performed in 3 distinct continuous movements per exercise: bilateral active exercise, alternating reciprocal movement, and a sustained contralateral hold.

Ten to 15 repetitions are performed for each movement successively, without rest, to complete 1 set. The goal is the ability to perform 2 full cycles of the entire program without pain, using sound technique and no substitution. Full description of the program is available.⁵²

APPENDIX D

RETURN TO THROWING

Criteria to Initiate Phase 1 Interval Throwing (Long Toss)

- Full, nonpainful ROM
 - Shoulder total ER/IR ROM in 90° of shoulder abduction within 5° of nonthrowing shoulder
 - Shoulder horizontal adduction of 40° or greater on throwing shoulder
 - Glenohumeral IR deficit of less than 15°
 - Elbow and wrist passive ROM within normal limits
- Shoulder, elbow, and wrist strength based on manual muscle test, handheld dynamometer, or isokinetic testing
 - ER/IR ratio of 72% to 76%
 - ER/abduction ratio of 68% to 73%
 - Throwing-shoulder IR greater than 115% compared to nonthrowing shoulder
 - Throwing-shoulder ER greater than 95% compared to nonthrowing shoulder
 - Throwing-arm elbow flexion/extension 100% to 115% compared to nonthrowing arm
 - Throwing-arm wrist flexion/extension and forearm pronation/supination 100% to 115% compared to nonthrowing arm
- Satisfactory clinical examination
 - No pain, tenderness, or effusion
 - Negative laxity testing: prone valgus stress and milking maneuver (**ONLINE VIDEO**)
 - Negative special test for other elbow or shoulder pathology
 - Physician and rehabilitation team clearance
- Successful completion of all steps in the rehabilitation process
- Satisfactory functional test scores
 - Prone ball-drop test (throwing side 110% or greater compared to the nonthrowing side) (**ONLINE VIDEO**)
 - One-arm ball throws against the wall using a 0.9 kg (2 lb) plyoball for 30 seconds without pain and exhibiting the ability to maintain 90°/90° arm position without compensation (throwing side greater than 90% of nonthrowing side) (**ONLINE VIDEO**)
 - Throwing into plyoback rebounder with 0.45-kg (1-lb) plyoball for 30 seconds with no pain, normal mechanics (without substitution), and good control
 - Single-leg step-down for 30 seconds, controlling pelvis and lower extremity alignment for both sides (limb symmetry: 95% or greater) (**ONLINE VIDEO**)
 - Prone plank test for time (**ONLINE VIDEO**)
- Minimum Kerlan-Jobe Orthopaedic Clinic throwers' assessment score of 85

Abbreviations: ER, external rotation; IR, internal rotation; ROM, range of motion.

Criteria are based on several sources.²